Examples

We will use MatCont to start homoclinic orbits that emanate from BT points in several models[1]. Recall that in the GUI of MatCont k is denoted as `TTolerance`. In all examples, we will set `TTolerance`$=10^{-5}$ (k). As a rule, the `amplitude` (A) should always be larger than `TTolerance` given the geometric meaning of both variables, *cf.* Figure 1. In all cases, ε_0 and ε_1 are the free homoclinic parameters since this appears to be the most stable choice. As another rule, the BT point itself should be computed to a geometric precision significantly smaller than `TTolerance`. This can be achieved in MatCont by decreasing the tolerances `VarTolerance` and `TestTolerance` for the curve on which the BT points are detected.

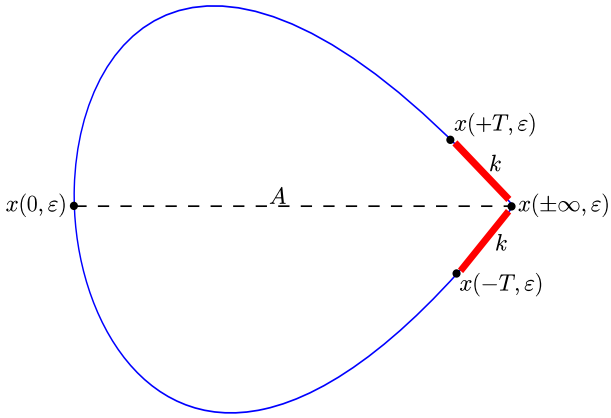


Figure 1: The initial homoclinic solution.

1 Morris-Lecar Model

Every cell in our body has a membrane that controls the movement of ions in and out of the cell. Cell membranes have specific channels that allow ions to move across the membrane.

In the Morris-Lecar model [10] we have calcium (Ca^{2+}) and potassium (K^{1+}) voltage channels and a leak (L) channel. The resulting system is

$$\begin{cases} C\dot{V} &= I_{app} - I_{ion}, \\ \dot{w} &= \frac{\phi(w_\infty - w)}{\tau}, \end{cases} \tag{1}$$

where

$$\begin{aligned} I_{ion} &= g_{Ca} m_\infty (V - V_{Ca}) + g_K w (V - V_K) + g_L (V - V_L), \\ m_\infty &= 0.5(1 + \tanh(\tfrac{V - v_1}{v_2})), \\ w_\infty &= 0.5(1 + \tanh(\tfrac{V - v_3}{v_4})), \end{aligned}$$

and

$$\tau = \frac{1}{\cosh(\frac{V - v_3}{2v_4})}.$$

In these equations, C is the membrane capacitance; I_{app} is the applied current; I_{ion} collects the Ca^{2+}, K^{1+} and L currents; g_L, g_{Ca} and g_K are the maximal conductances for L, Ca^{2+} and K^{1+} channels, respectively. V is the membrane potential; V_L, V_{Ca} and V_K are the equilibrium potentials corresponding to L,Ca^{2+} and K^{1+} conductance's, respectively; w is the fraction of open K^{1+} channels; m_∞ and w_∞ are the fractions of open Ca^{2+} and K^{1+} channels at steady state respectively; $\frac{\tau}{\phi}$ determines the activation

[1] The examples require MatCont 6.2 or higher

time for the K^{1+} current. Note that v_1,v_2,v_3 and v_4 are parameters chosen to fit the model data. Fix the parameter values as follows, $C = 20$, $V_L = -60$, $V_{Ca} = 120$, $V_K = -84$, $g_L = 2$, $g_{Ca} = 4.4$, $g_K = 8$, $v_1 = -1.2$, $v_2 = 18$, $v_3 = 2$, $v_4 = 30$, $\phi = \frac{1}{25}$. Then compute the equilibrium curve with free parameter I_{app}, starting with initial values $I_{app} = 0$, $V = -60.85456779$ and $w = 0.0149139691$. Two Hopf (H) points are detected. In the **Continuer** window set `MaxStepSize` = 1. Start from any H point and compute the Hopf curve passing through it with (I_{app}, v_3) free parameters. Four BT points are detected, see Table 1 and 2. Compute the homoclinic to saddle curve from any BT point with: (I_{app}, v_3) free system parameters, (`eps0`,`eps1`) as free homoclinic parameters, `TTolerance` $= 1 \times 10^{-5}$, `ntst` = 80, `Adapt` = 1 and initial numerical data as in Table 3, see Figure 2.

Label	I_{app}	v_3	**State variables**
$\mathbf{BT}_1$	519.63	−63.40	$(-11.15, 0.97)$
$\mathbf{BT}_2$	488.00	−51.78	$(2.82, 0.97)$
$\mathbf{BT}_3$	−227.13	69.76	$(13.61, 0.02)$
$\mathbf{BT}_4$	48.23	21.73	$(-27.15, 0.04)$

Table 1: Parameter, state values at the bifurcation points in Figure 2.

Label	a	b	d	e	a_1	b_1
$\mathbf{BT}_1$	7.4×10^{-4}	2.3×10^{-2}	-7.5×10^{-6}	-10×10^{-4}	-2.5×10^{-3}	-5.3×10^{-2}
$\mathbf{BT}_2$	-7.5×10^{-4}	-3.0×10^{-2}	-4.0×10^{-5}	-1.4×10^{-3}	5.2×10^{-3}	7.1×10^{-2}
$\mathbf{BT}_3$	-1.1×10^{-3}	-3.3×10^{-2}	2.6×10^{-5}	9.5×10^{-4}	-2.2×10^{-4}	-5.8×10^{-2}
$\mathbf{BT}_4$	2.8×10^{-4}	1.5×10^{-2}	4.4×10^{-6}	4.7×10^{-4}	-5.5×10^{-4}	6.6×10^{-2}

Table 2: BT normal form coefficient $\{a, b, d, e, a_1, b_1\}$ values at the bifurcation points in Figure 2.

Label	`amplitude`	**Initial** T	**Initial** ε	**Compute|**
$\mathbf{BT}_1$	3.2×10^{-5}	22046.73	6.28×10^{-5}	**Backward**
$\mathbf{BT}_2$	1.4×10^{-5}	10993.52	4.18×10^{-5}	**Forward**
$\mathbf{BT}_3$	1.4×10^{-5}	8988.07	5.11×10^{-5}	**Backward**
$\mathbf{BT}_4$	3.1×10^{-5}	35732.42	3.78×10^{-5}	**Forward**

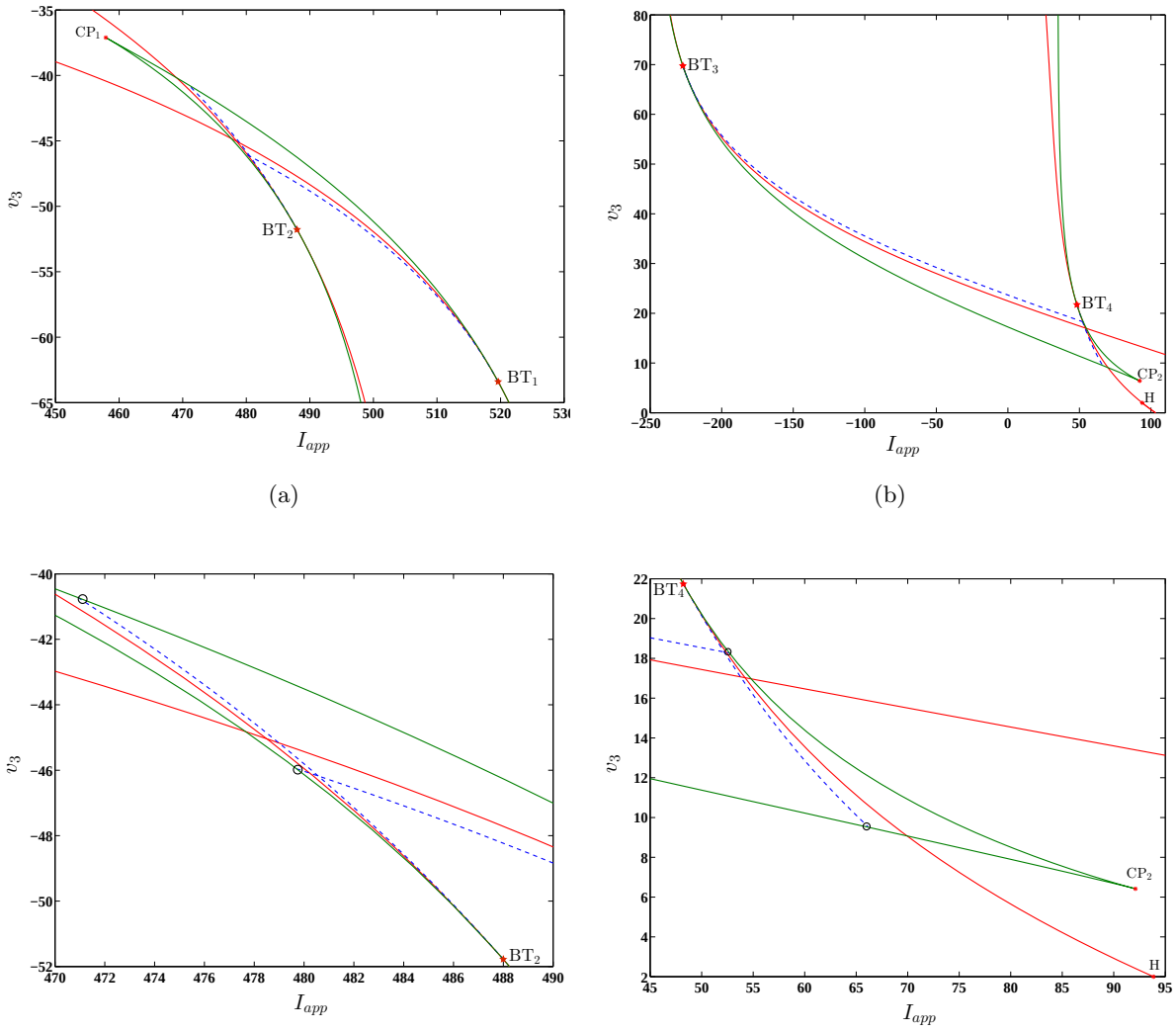
Table 3: The initial homoclinic continuation data at each BT point defined as in Table.1.

2 Predator-prey systems with constant rate harvesting

Consider the following predator-prey system with constant rate predator harvesting [4]:

$$\begin{cases} \dot{x} = rx(1 - \frac{x}{k}) - y\frac{x}{e+x}, \\ \dot{y} = y(-d + \frac{x}{e+x}) - h, \end{cases} \tag{2}$$

where k is the carrying capacity of the prey population, d is the death rate of the predator, r is the intrinsic growth rate of the prey population, and h is the harvesting rate. The function $\frac{x}{e+x}$ is often called the functional response of Holling type II. Xiao and Ruan [15] show the existence of a BT bifurcation in (2) and sketch the global bifurcation diagram including the homoclinic curve which emanates from the BT point. We study the occurrence of homoclinic orbits that emanate from the computed BT point using MatCont. We fix the parameter values as follows: $r = 1$, $e = 1$, $k = 2$, $h = 0.5$, then we compute the equilibria with free parameter d (initially $d = 0$) and initial value for state variables $x = 1.1$ and $y = 0.09$. A limit point (LP) is detected. Compute the fold curve passing through it with (d, h) as free parameters. One BT point is detected (see Table 4). To initialize the homoclinic orbit from the BT point with (d, h) as free system parameters input the following numerical data in the MatCont **Starter** window: `amplitude` $= 1 \times 10^{-3}$, `TTolerance` $= 1 \times 10^{-5}$ and `ntst` = 40. In the **Continuer** window set `Adapt` = 1. Activate `eps0` and `eps1` as free homoclinic parameters and then **Compute|Backward**. The result is shown in Figure 4. The initial ε is 5.475×10^{-3} and during continuation $T = 856.934$.



(a) (b)

(c) (d)

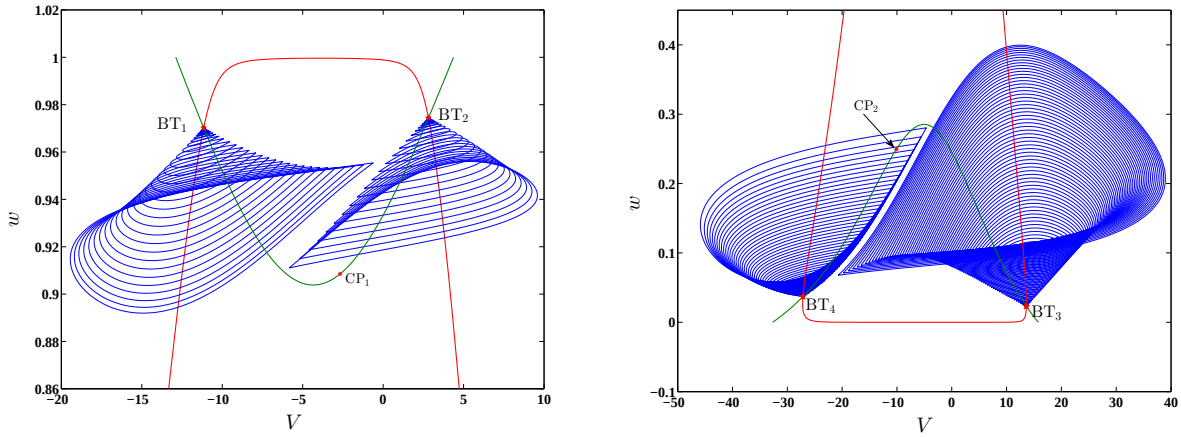
Figure 2: (a) and (b) Homoclinic orbits in parameter space (I_{app}, v_3). The dashed blue curves are homoclinic curves. The green line is the LP curve. The red line is the Hopf curve. (c) and (d) The same curves as in Figure 2a and Figure 2b, respectively, zoomed near the $\mathrm{BT}_{2,4}$ points. The non-central-homoclinic-to-saddle-node orbit on the LP curve is circled, where the homoclinic curve ends.

3 CO-oxidation in a platinum model.

Consider the following chemical model which describes CO-oxidation in platinum [7, 5, 3]

$$\begin{cases} z = 1 - x - y - s, \\ \dot{x} = 2k_1 z^2 - 2k_{-1}x^2 - k_3 xy, \\ \dot{y} = k_2 z - k_{-2} y - k_3 xy, \\ \dot{s} = k_4(z - \lambda s). \end{cases} \tag{3}$$

where $\lambda = \frac{k_{-4}}{k_4}$. The underlying reaction scheme is studied in [5] and we notice that a factor 2 is missing in front of $k_1 z^2$ in [7]. The reaction rates (constants) above are fixed as follows $k_1 = 2.5$, $k_{-1} = 1$, $k_3 = 10$, $k_{-2} = 0.1$, $k_4 = 0.0675$, $k_2 = 1.4707$, $\lambda = 0.4$. Compute the equilibria curve with free k_2 and initial state variables $x = 0.00295$, $y = 0.76211$, $s = 0.1678$. Two limit points (LP) are detected. Start the LP continuation from any of the two LP-points with (λ, k_2) free system parameters. Two

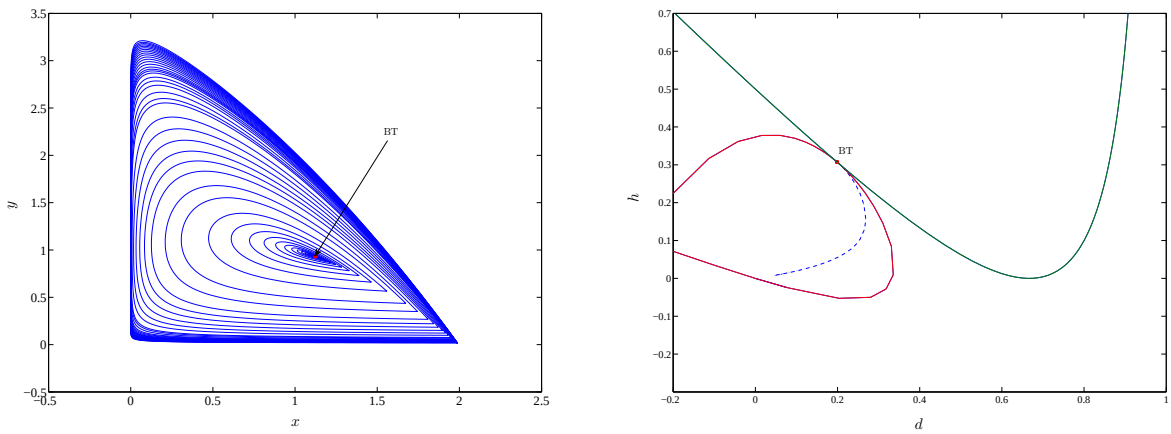


(a) (b)

Figure 3: Homoclinic orbits (blue curves) in the Morris-Lecar model in state space (V, w). The green line is the LP curve; the red line is the Hopf curve.

Label	$\boldsymbol{d}$	$\boldsymbol{h}$	**State variables**	$\boldsymbol{a}$	$\boldsymbol{b}$	$\boldsymbol{d}$	$\boldsymbol{e}$	$\boldsymbol{a_1}$	$\boldsymbol{b_1}$
BT	0.2	0.31	$(1.12, 0.93)$	-0.18	0.38	0.06	-0.36	0.17	0.16

Table 4: Parameter, state and BT normal form $\{a,\, b,\, d,\, e,\, a_1,\, b_1\}$ coefficients at the bifurcation point in Figure 4.



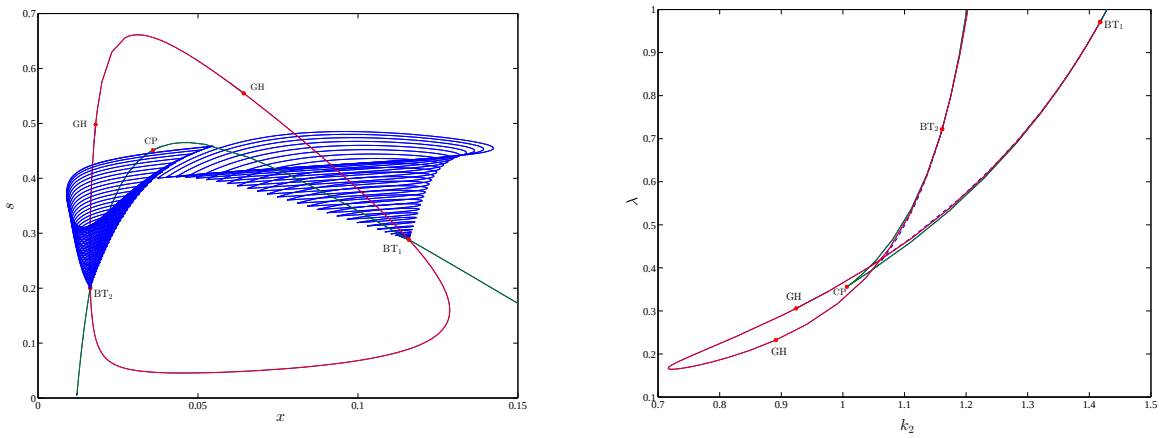
(a) (b)

Figure 4: (a) The homoclinic orbits in state space for system (2), (b) The homoclinic orbits in parameter space. The dashed blue curve is the homoclinic curve. The red is the Hopf curve and the green is the LP curve.

BT points are detected, see Table 5. From BT_1 and with $(k_2,\, \lambda)$ as free system parameters start the homoclinic curve continuation using `amplitude` $= 2 \times 10^{-5}$, `TTolerance` $= 1 \times 10^{-5}$ and `ntst` $= 40$. In the **Continuer** window set `InitStepsize` $= 1 \times 10^{-3}$ and `Adapt` $= 1$. Then **Compute**|**Backward** with (`eps0`,`eps1`) as free homoclinic parameters. It turns out that the initial ε is 5.28×10^{-4} and $T = 1667.78$. For BT_2 set `InitStepsize` $= 1 \times 10^{-2}$, use `amplitude` $= 1.2 \times 10^{-6}$, `TTolerance` $= 1 \times 10^{-8}$, `ntst` $= 40$, `Adapt` $= 1$ then **Compute**|**Backward**. Notice that the initial ε is 9.821×10^{-5} and during continuation $T = 49559.486$. The results are presented in Figure 5

Label	k_2	λ	State variables	a	b	d	e	a_1	b_1
$\mathbf{BT}_1$	1.42	0.97	$(0.12, 0.32, 0.29)$	-0.08	-2.14	-0.18	-8.25	-0.86	-7.12
$\mathbf{BT}_2$	1.161	0.72	$(0.02, 0.64, 0.20)$	-0.05	-1.94	-0.07	-8.14	0.66	13.45

Table 5: Parameter, state and BT normal form coefficient $\{a,\ b,\ d,\ e,\ a_1,\ b_1\}$ values at the BT points in Figure 5.



(a) (b)

Figure 5: (a) Homoclinic orbits in (x, s)-space for the CO−oxidation model, (c) Homoclinic orbits in parameter space. The dashed blue curve is the homoclinic curve. The green curve is the LP curve and the red is the Hopf curve.

4 Indirect field oriented control system

The indirect field oriented control (IFOC) system of an induction motor can be mathematically described as in [1, 11] by the following ODEs:

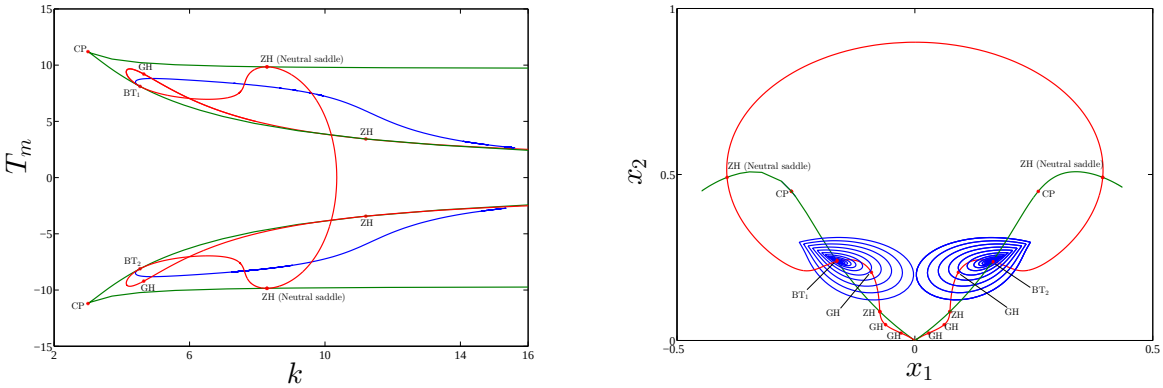
$$\begin{cases} \dot{x}_1 = -c_1 x_1 + c_2 x_4 - \frac{k c_1}{u_2^0} x_2 x_4, \\ \dot{x}_2 = -c_1 x_2 + c_2 u_2^0 + \frac{k c_1}{u_2^0} x_1 x_4, \\ \dot{x}_3 = -c_3 x_3 - c_4 c_5 \left(x_2 x_4 - u_2^0 x_1\right) + \left(c_4 T_m + c_3 w_{ref}\right), \\ \dot{x}_4 = -(k_i - k_p c_3) x_3 - k_p c_4 c_5 \left(x_2 x_4 - u_2^0 x_1\right) + k_p \left(c_4 T_m + c_3 w_{ref}\right). \end{cases} \tag{4}$$

Here x_1, x_2, x_3 and x_4 are the state variables, where x_1 and x_2 denote direct and quadrature components of the rotor flux; x_3 is the rotor speed error (i.e., the difference between the reference and the real mechanical speeds of the rotor); and x_4 denotes the quadrature axis component of the stator current. We also define the following constants and parameters: u_2^0 is a constant reference for the rotor flux magnitude; c_1 to c_5 are machine parameters; k_p and k_i are the proportional (P) and the integral (I) control gains, respectively; w_{ref} is the speed reference; T_m the load torque; k the measure of rotor time constant mismatches. The occurrence of LP and Hopf in IFOC has been characterized as a result of rotor time constant mismatches (see, for example, [2, 6] and [9]). The first results on the occurrence of a BT bifurcation in the IFOC model were presented in [12]. A detailed analytical study for the codim 2 bifurcations of (4) can be found in [11]. By continuation of equilibria with free k (initially $k = 17$) and fixed parameters $T_m = 5$, $c_1 = 4.4868$, $c_2 = 0.3567$, $c_3 = 0$, $c_4 = 9.743$, $c_5 = 1.911$, $u_2^0 = 11.3$, $k_p = 4.5$, $k_i = 500$, $w_{ref} = 0$, and initial point $(x_1, x_2, x_3, x_4) = (-0.21, 0.11, 2.5, 2)$, MatCont detects an LP point and a Hopf point. Further, by continuation of the LP with (k, T_m) free, a BT point is detected (in Table 6 this point is labeled by BT_2). Select BT_2 and compute the the Hopf curve passing through it with (k, T_m) free, an extra BT point is detected (labeled by BT_1). From the BT_1 point we start the continuation of the homoclinic curve, using k and T_m as free system parameters, (`eps0`,`eps1`) as free homoclinic parameters, initial `amplitude` $= 2 \times 10^{-5}$, `TTolerance` $= 1 \times 10^{-5}$, `ntst` $= 40$, `Adapt`

Label	k	T_m	State variables	a	b	d	e	a_1	b_1
BT$_2$	4.54	8.11	$(-0.16, 0.24, 0.00, 10.06)$	28.01	-0.91	-4.47	-0.31	4.39	0.07
BT$_1$	4.54	-8.11	$(0.16, 0.24, 0.00, -10.06)$	28.01	-0.91	-4.47	-0.31	4.39	0.07

Table 6: Parameter, state and BT normal for coefficient $\{a, b, d, e, a_1, b_1\}$ values at the bifurcation points in Figure 6.

$= 1$. Then **Compute**|**Backward**, noticing that during continuation $T = 91.221$. For BT$_2$ the same procedure then **Compute**|**Forward** can be used, noticing that during continuation $T = 91.221$. In both cases, the initial ε is 0.01 See Figure 6.



(a) (b)

Figure 6: (a) Homoclinic orbits emanating from the BT points of the IFOC model in parameter space (k, T_m). The blue dashed curve is the homoclinic orbit. The black is the equilibria curve, the green is the LP curve and the red is the Hopf curve. (b) The homoclinic orbits in state space (x_1, x_2).

5 The extended Lorenz-84 model

This example is an extended version of the Lorenz-84 model. In this model we can find all codim 2 points of equilibria (i.e., BT, CP, GH, ZH and HH) [8, 14, 13]. Here, we discuss the switching from the BT points to the homoclinic branches.

The extended Lorenz-84 model has the form

$$\begin{cases} \dot{X} = -Y^2 - Z^2 - \alpha X + \alpha F - \xi U^2, \\ \dot{Y} = XY - \beta XZ - Y + G, \\ \dot{Z} = \beta XY + XZ - Z, \\ \dot{U} = -\delta U + \xi UX + S. \end{cases} \tag{5}$$

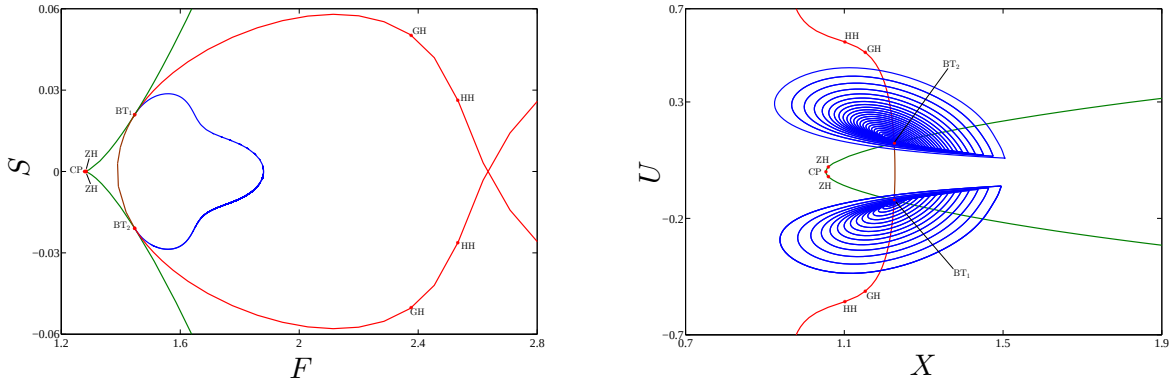
In this system, X models the intensity of a baroclinic wave, Y and Z the sin and cos coefficients of the wave respectively, the variable U is added to study the influence of external parameters such as temperature. The parameters are fixed as follows $\alpha = 0.25$, $\beta = 1$, $G = 0.25$, $\delta = 1.04$, $\xi = 0.987$ and $F = 2.61$. Start the equilibria continuation with free S (initially $S = 0$) and with initial state variable values $X = 1.05$, $Y = -0.01$, $Z = 0.21$, $U = -0.5$. Two LP points are detected. From any LP point compute the LP curve passing through it with (F, S) as free system parameters. Two BT points are found (See Table 7).

From BT$_1$ start the homoclinic continuation with (F, S) free, `amplitude` $= 4 \times 10^{-5}$, `TTolerance` $= 1 \times 10^{-5}$, `ntst` $= 40$, `Adapt` $= 3$ and choose (`eps0`, `eps1`) as the free homoclinic parameters then **Compute**|**Forward**. In the resulting continuation $T = 1346.112$ is fixed and the initial ε is 1.2×10^{-3}.

Label	F	S	State variables	a	b	d	e	a_1	b_1
$\mathbf{BT}_1$	1.45	0.02	$(1.23, -0.04, 0.20, -0.12)$	-0.21	-0.61	-0.24	-2.52	-0.59	-1.24
$\mathbf{BT}_2$	1.45	-0.02	$(1.23, -0.04, 0.20, 0.12)$	0.21	0.61	-0.24	-2.52	0.59	1.24

Table 7: Parameter, state and BT normal form coefficient $\{a, b, d, e, a_1, b_1\}$ values at the bifurcation points in Figure 7.

The obtained homoclinic orbits are as in Figure 7. For BT_2 the same procedure works with `amplitude` $= 4 \times 10^{-5}$ and then **Compute|Backward** leads to a continuation with $T = 1346.112$ as well. The initial ε is 1.2×10^{-3}.



(a) (b)

Figure 7: (a) Homoclinic orbits in parameter space. The blue curve is the homoclinic curve. The red is the Hopf curve and the green is the LP curve, (b) Homoclinic orbits in $(X, U)-$space for the Lorenz-84 model.

Conclusions We have described an initializer implemented in MatCont to start homoclinic orbits from a BT point based on the Lindstedt-Poincaré perturbation method. The initializer allows to compute the initial homoclinic solution and parameters so that the continuation of the homoclinic orbit can be started. We suggest to allow `eps0` and `eps1` to vary as homoclinic parameters and to set `TTolerance` $= 1 \times 10^{-5}$. In the MatCont **continuer** window set **Adapt** $= 1$. Then start to increase/decrease the `amplitude` value. This works for most studied models. However, this choice is not an absolute rule and it takes some trial-and-error to set all parameters (including `TTolerance`, continuation parameters and adaptation (`Adapt`)) for the successful continuation. Note that in each case both **Compute|Forward** and **Compute|Backward** should be tried. In general, cases where a has small (absolute) value are more difficult to handle than cases where a is moderate.

References

[1] A. S. Bazanella and R. Reginatto. Robustness margins for indirect field oriented control of induction motors. *IEEE Trans. Autom. Contr.*, 45(6):1226–1231, 2000.

[2] A. S. Bazanella and R. Reginatto. Instabilty mechanisms in indirect field oriented control drives: Theory and experimental results. In *IFAC World Congress*, volume 15, Part I, pages 1572–1572, Barcelona, Spain, 2002.

[3] W.-J. Beyn. Numerical analysis of homoclinic orbits emanating from a Takens-Bogdanov point. *IMA J. Numer. Anal.*, 14(3):381–410, 1994.

[4] F. Brauer and A. C. Soudack. Stability regions and transition phenomena for harvested predator-prey systems. *J. Math. Biol*, 7:319–337, 1979.

[5] V. I. Bykov, G. S. Yablonskii, and V. F. Kim. On one simple model of kinetic self-oscillations in catalytic reaction of CO-oxidation. *Dokl. Sov. Math.*, pages 637–639, 1978.

[6] F. Gordillo, F. Salas, R. Ortega, and J. Aracil. Hopf bifurcation in indirect field-oriented control of induction motors. *Automatica*, 38:829–835, 2002.

[7] A. I. Khibnik. LINLBF: A program for continuation and bifurcation analysis of equilibria up to codimenson three. In D. Roose, B. De Dier, and A. Spence, editors, *Continuation and Bifurcations: Numerical Techniques and Applicatins*, NATO ASI Series C313, pages 283–296. Kluwer Academic, 1990.

[8] Yu. A. Kuznetsov, H.G.E. Meijer, W. Govaerts, and B. Sautois. Switching to nonhyperbolic cycles from codim 2 bifurcations of equilibria in odes. *Physica D*, 237:3061–3068, 2008.

[9] J. Moiola and G. Chen. *Hopf Bifurcation Analysis: A Frequency Domain Approach.* World Scientific, Singapore, 1996.

[10] C. Morris and H. Lacar. Voltage oscillations in the barnacle giant muscle fiber. *Biophys. J.*, 35:1418–1450, 1991.

[11] F. Salas, F. Gordillo, and J. Aracil. Codimension-two bifurcations in indirect field oriented control of induction motor drives. *Internat. J. Bifur. Chaos*, 18(3):779–792, 2008.

[12] F. Salas, R. Reginatto, F. Gordillo, and J. Aracil. Bogdanov-Takens bifurcation in indirect field oriented control of induction motor drives. In *Proc. CDC*, pages 4357–4362, Atlantis, Bahamas, 2004.

[13] A. Shil'nikov, G. Nicolis, and C. Nicolis. Bifurcation and predictability analysis of a low-order atmospheric circulation model. *Internat. J. Bifur. Chaos*, 5(6):1701–1711, 1995.

[14] Lennaert van Veen. Baroclinic flow and the Lorenz-84 model. *Internat. J. Bifur. Chaos*, 13(8):2117–2139, 2003.

[15] D. Xiao and S. Ruan. Bogdanov-takens bifurcations in harvested predator-prey systems. *Fields Institute Communications*, 21:493–506, 1999.